

SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 0605

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

***C02:** Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks:25

Annexure A

Analytical Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Q11. Perceptron Model

Question

A perceptron model is used to classify whether a customer will purchase a product based on two features: advertisement exposure and income level. The weights are **0.5** and **0.3**, and the bias is **-0.4**. For a customer with advertisement exposure = **2** and income level = **1**, compute the net input to the perceptron and apply the step activation function. Determine whether the customer makes a purchase or not.

Given

- Advertisement Exposure (x_1) = 2
 - Income Level (x_2) = 1
 - Weight₁ (w_1) = 0.5
 - Weight₂ (w_2) = 0.3
 - Bias (b) = -0.4
-

Formula

$$Net = w_1x_1 + w_2x_2 + b$$

Calculation

$$\begin{aligned} Net &= (0.5 \times 2) + (0.3 \times 1) - 0.4 \\ &= 1 + 0.3 - 0.4 \\ &= 0.9 \end{aligned}$$

Step Activation Function

$$Output = \begin{cases} 1, & Net \geq 0 \\ 0, & Net < 0 \end{cases}$$

Since

$$0.9 > 0$$

Output

$$= 1$$

Final Answer

Net Input = 0.9

Activation Output = 1

Interpretation

Since the output is **1**, the perceptron predicts that **the customer will purchase the product.**

Q12. Delta Learning Rule (5 Marks)**Given**

Initial Weight

$$w = 0.6$$

Learning Rate

$$\eta = 0.15$$

Input

$$x = 2.5$$

Actual Output

$$y = 0.7$$

Target Output

$$t = 1$$

Step 1**Error**

Formula

$$Error = t - y$$

Calculation

$$\begin{aligned} &1 - 0.7 \\ &= 0.3 \end{aligned}$$

Step 2

Weight Update

Formula

$$\Delta w = \eta \times Error \times x$$

Substitute

$$\begin{aligned} &= 0.15 \times 0.3 \times 2.5 \\ &= 0.1125 \end{aligned}$$

Step 3

Updated Weight

Formula

$$w_{new} = w + \Delta w$$

Calculation

$$\begin{aligned} &0.6 + 0.1125 \\ &= 0.7125 \end{aligned}$$

Final Answer

Error

$$0.3$$

Weight Update

$$0.1125$$

Updated Weight

$$0.7125$$

Interpretation

The neuron increases its weight because the predicted output is lower than the target output.

Q13. Confusion Matrix Analysis (5 Marks)

Given

$$TP = 140$$

$$TN = 180$$

$$FP = 25$$

$$FN = 35$$

Step 1

Accuracy

Formula

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Calculation

$$\begin{aligned} &= \frac{140 + 180}{140 + 180 + 25 + 35} \\ &= \frac{320}{380} \\ &= 0.8421 \end{aligned}$$

Accuracy

$$84.21\%$$

Step 2

Precision

Formula

$$\begin{aligned} Precision &= \frac{TP}{TP + FP} \\ &= \frac{140}{165} \\ &= 84.85\% \end{aligned}$$

Step 3

Recall

$$\begin{aligned} \text{Recall} &= \frac{TP}{TP + FN} \\ &= \frac{140}{175} \\ &= 80\% \end{aligned}$$

Step 4

Specificity

$$\begin{aligned} \text{Specificity} &= \frac{TN}{TN + FP} \\ &= \frac{180}{205} \\ &= 87.80\% \end{aligned}$$

Step 5

F1 Score

Formula

$$F1 = \frac{2PR}{P + R}$$

Substitute

$$\begin{aligned} &= \frac{2(0.8485)(0.80)}{0.8485 + 0.80} \\ &= 0.8236 \end{aligned}$$

F1 Score

$$82.36\%$$

Final Answer

Metric	Value
Accuracy	84.21%
Precision	84.85%
Recall	80.00%
Specificity	87.80%
F1 Score	82.36%

Analysis

The model has **high precision (84.85%)**, meaning few false positives are produced.

Since fraud detection aims to minimize false alarms, the model is **reasonably suitable**. However, the recall of **80%** indicates that some fraud cases are still missed. Improving recall while maintaining precision would make the model more effective for fraud detection.

Q14. Genetic Algorithm (5 Marks)

Given

$$f(x) = 3x + 2$$

Population

$$\{2, 5, 7, 9\}$$

Step 1

Fitness Calculation

For

$$\begin{aligned}x &= 2 \\f(2) &= 3(2) + 2 = 8\end{aligned}$$

For

$$\begin{aligned}x &= 5 \\f(5) &= 17\end{aligned}$$

For

$$x = 7$$
$$f(7) = 23$$

For

$$x = 9$$
$$f(9) = 29$$

Fitness Table

Individual Fitness

2	8
5	17
7	23
9	29

Best Candidates

Highest fitness

9

Second Highest

7

Crossover

Parent 1

1001

Parent 2

0111

Single-point crossover

100|1

011|1

Offspring

1001

0111

(You can mention that different crossover points would generate different offspring; this is just one illustration.)

Mutation

Example

1001

↓

1011

One bit changes randomly.

Mutation helps

- Maintain diversity
- Prevent premature convergence
- Escape local optimum
- Improve exploration

Final Answer

The best chromosomes are

9

and

7

because they have the highest fitness values.

Mutation introduces new genetic variations and helps the Genetic Algorithm avoid getting stuck in local optima.

Q15. Deep Learning Model Analysis (5 Marks)

Given

Training Accuracy

65%

↓

97%

Validation Accuracy

63%

↓

75%

↓

72%

Identify the Problem

The model is suffering from

Overfitting

Why?

Training accuracy keeps increasing.

Validation accuracy stops improving after

75%

and starts decreasing.

This indicates

The model memorizes training data instead of learning general patterns.

Techniques to Reduce Overfitting

1. Dropout

Randomly deactivates neurons during training.

Benefits

- Prevents dependency on specific neurons
 - Improves generalization
 - Reduces overfitting
-

2. Data Augmentation

Creates additional training samples by applying transformations such as rotation, flipping, cropping, or scaling.

Benefits

- Increases dataset diversity
 - Improves robustness
 - Reduces overfitting
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Importance of Validation Metrics

Validation metrics help evaluate how well the model performs on unseen data.

They are used to:

- Detect overfitting and underfitting
 - Compare different models
 - Tune hyperparameters
 - Select the best-performing model
 - Ensure good generalization
-

Final Answer

The model is experiencing **overfitting** because the training accuracy continues to improve while the validation accuracy decreases after reaching its peak. Techniques such as **Dropout** and **Data Augmentation** can reduce overfitting by improving the model's ability to generalize. Validation metrics are essential because they evaluate the model's performance on unseen data and help in selecting a reliable model for real-world applications.